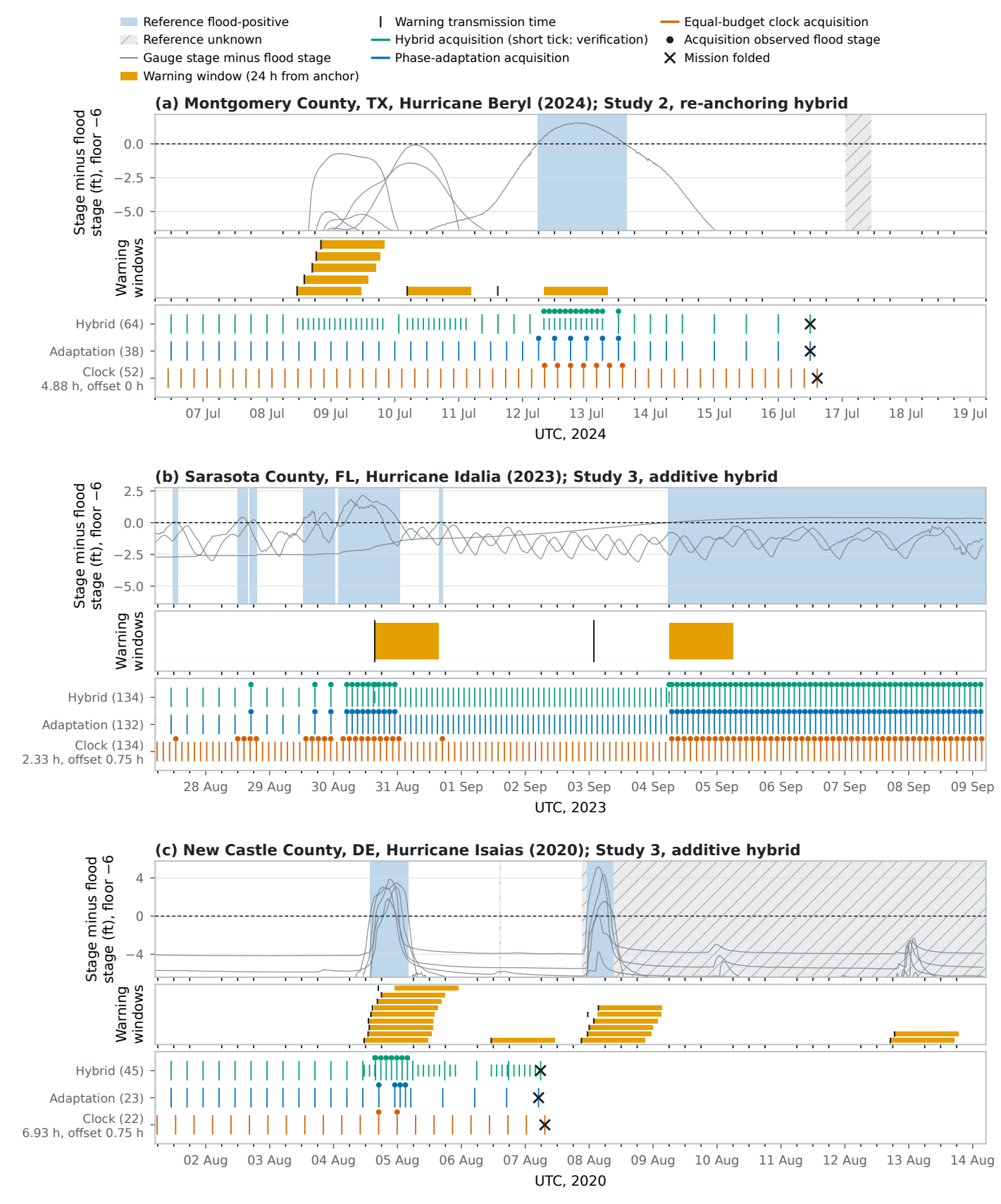

Reference flood-positive
Reference unknown
Gauge stage minus flood stage
Warning window (24 h from anchor)
Warning transmission time
Hybrid acquisition (short tick: verification)
Phase-adaptation acquisition
Equal-budget clock acquisition
Acquisition observed flood stage
Mission folded
(a) Montgomery County, TX, Hurricane Beryl (2024); Study 2, re-anchoring hybrid
Stage minus flood stage (ft), floor −6
0.0
−2.5
−5.0
Warning windows
Hybrid (64)
Adaptation (38)
Clock (52)
4.88 h, offset 0 h
07 Jul
08 Jul
09 Jul
10 Jul
11 Jul
12 Jul
13 Jul
14 Jul
15 Jul
16 Jul
17 Jul
18 Jul
19 Jul
UTC, 2024
(b) Sarasota County, FL, Hurricane Idalia (2023); Study 3, additive hybrid
Stage minus flood stage (ft), floor −6
2.5
0.0
−2.5
−5.0
Warning windows
Hybrid (134)
Adaptation (132)
Clock (134)
2.33 h, offset 0.75 h
28 Aug
29 Aug
30 Aug
31 Aug
01 Sep
02 Sep
03 Sep
04 Sep
05 Sep
06 Sep
07 Sep
08 Sep
09 Sep
UTC, 2023
(c) New Castle County, DE, Hurricane Isaias (2020); Study 3, additive hybrid
Stage minus flood stage (ft), floor −6
4
0
−4
Warning windows
Hybrid (45)
Adaptation (23)
Clock (22)
6.93 h, offset 0.75 h
02 Aug
03 Aug
04 Aug
05 Aug
06 Aug
07 Aug
08 Aug
09 Aug
10 Aug
11 Aug
12 Aug
13 Aug
14 Aug
UTC, 2020